

COSC3506/ITEC3506

Software Engineering

Module 8 Lesson 1

An Example of OO Analysis & Design

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OO Analysis & Design

-----Sequence of activities

- Identify a first set of candidate classes
- Add associations and attributes
- Find generalizations
-
- List the main **responsibilities** of each class
- Decide on specific operations
- Iterate over the entire process until the model is satisfactory
 - Add or delete classes, associations, attributes, generalizations, responsibilities or operations
- Don't be too disorganized.

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A simple technique for discovering domain classes – **Noun Extraction**

- Look at a source material such as a description of requirements & use cases
- Extract the nouns and noun phrases
- Eliminate nouns that:
 - are redundant
 - represent instances

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Object Oriented Analysis & Design

A system generates weather maps using data collected from unattended weather stations. Each weather station collects meteorological data and produces summaries of the data. On request, it sends the summary information to an area computer. The area computer uses a database of digitized maps to generate a set of local weather maps.

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Noun Identification

A **system** generates **weather maps** using **data** collected from unattended **weather stations**. Each weather station collects **meteorological data** and produces **summaries of the data**. On request, it sends the summary information to an **area computer**. The area computer uses a **database** of **digitized maps** to generate a set of local weather maps.

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Candidate Classes

System	general term
WeatherMap	
Data	same as MeteorologicalData
WeatherStation	<i>is this a general term?</i>
MeteorologicalData	<i>how does this relate to WeatherStation?</i>
DataSummary	<i>how does this relate to DataSummary?</i>
AreaComputer	hardware
Database	general term
DigitizedMap	

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Observations about the Candidate Classes

WeatherMap	is a DigitizedMap is derived from 1...* DataSummary
WeatherStation	has a set of MeteorologicalData
MeteorologicalData	
DataSummary	is derived from MeteorologicalData
DigitizedMap	

Can Meteorological Data be an attribute of WeatherStation?
Can DataSummary be combined with WeatherMap?

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Identifying attributes

- Look for information that must be maintained about each class
- Several nouns rejected as classes, may now become attributes
- An attribute should generally contain a simple value
 - e.g. string, number

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Attributes

WeatherStation
location metereologicalData

DigitizedMap
location geographicData

WeatherMap
location date-time geographicData weather

*Or should
MetereologicalData be a
separate object?*

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Tips about identifying and specifying valid associations

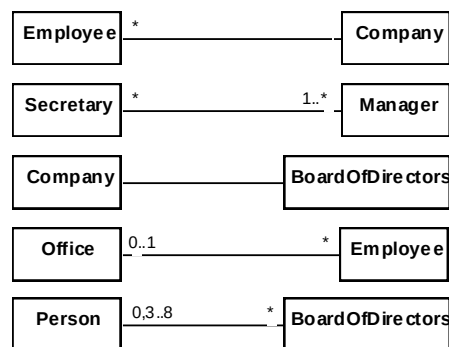
- An association should exist if a class
 - *possesses*
 - *controls*
 - *is connected to*
 - *is related to*
 - *is a part of*
 - *has as parts*
 - *is a member of, or*
 - *has as members*
 some other class in your model
- Specify the multiplicity at both ends
- Label it clearly.

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Associations and Multiplicity

- An association is used to show how two classes are related to each other
 - Symbols indicating multiplicity are shown at each end of the association

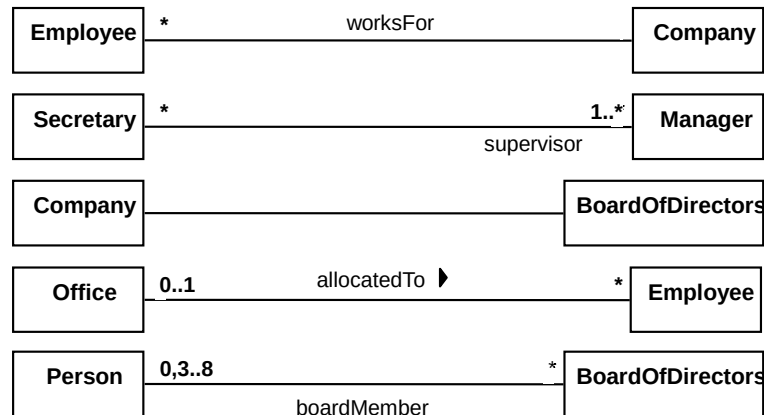


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Labelling associations

- Each association can be labelled, to make explicit the nature of the association

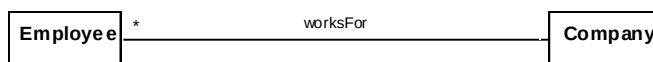


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Analyzing and validating associations

- **Many-to-one**
 - A company has many employees,
 - An employee can only work for one company.
 - This company will not store data about the moonlighting activities of employees!
 - A company can have zero employees
 - e.g. a 'shell' company
 - It is not possible to be an employee unless you work for a company



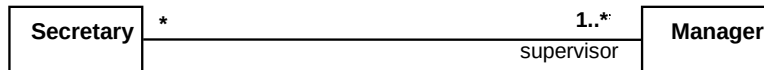
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Analyzing and validating associations

- **Many-to-many**

- A secretary can work for many managers
- A manager can have many secretaries
- Secretaries can work in pools
- Managers can have a group of secretaries
- Some managers might have zero secretaries.
- Is it possible for a secretary to have, perhaps temporarily, zero managers?



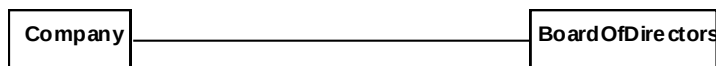
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Analyzing and validating associations

- **One-to-one**

- For each company, there is exactly one board of directors
- A board is the board of only one company
- A company must always have a board
- A board must always be of some company



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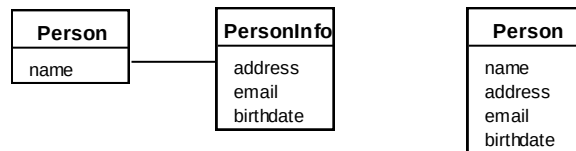
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Analyzing and validating associations

- Avoid unnecessary one-to-one associations

Avoid this

do this

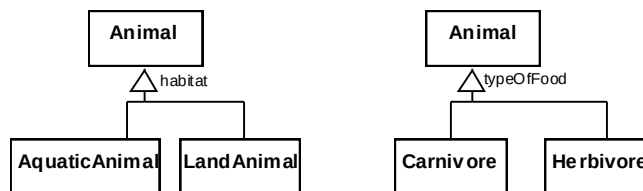


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Generalization

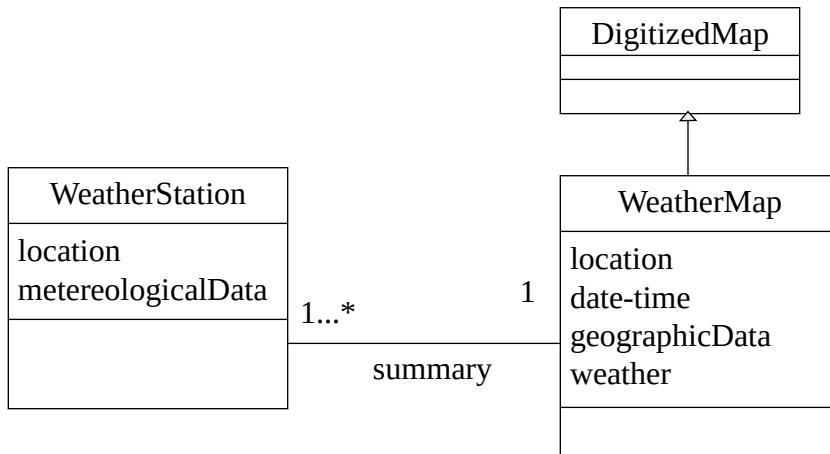
- Specializing a superclass into two or more subclasses
 - The *discriminator* is a label that describes the criteria used in the specialization



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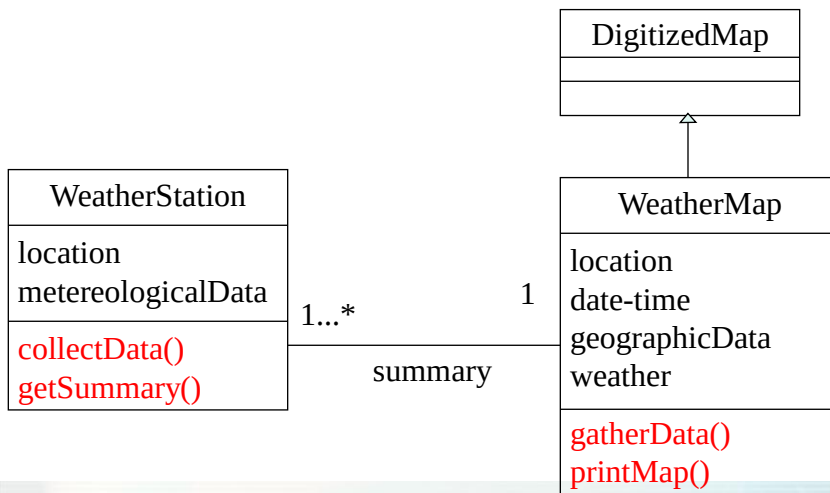
Class Diagram



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Class Diagram



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